

Circulating insulin-like growth factor (IGF) peptides and prostate cancer risk: a systematic review and meta-analysis

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Insulin-like growth factors (IGF-I, IGF-II) and their binding proteins (IGFBP-1-6) play a key role in cell proliferation, differentiation and apoptosis, suggesting possible involvement in carcinogenesis. Several epidemiological studies show associations of IGFs with prostate cancer. We searched the published literature for all studies relating levels of IGFs or IGFBPs with prostate cancer. We performed random effects meta-analysis to calculate summary odds ratios. The number of studies (prostate cancer cases) included in each meta-analysis were 42 (7,481) IGF-I; 10 (923) IGF-II; 3 (485) IGFBP-1; 5 (577) IGFBP-2; 29 (6,541) IGFBP-3; and 11 (3,545) IGF-I:IGFBP-3 ratio. The pooled odds ratios (95% confidence intervals) per standard deviation increase in peptide, were: IGF-I, OR = 1.21 (1.07, 1.36); IGF-II, OR = 1.17 (0.93, 1.47); IGFBP-1, OR = 1.21 (0.62, 2.33); IGFBP-2, OR = 1.18 (0.90, 1.54); IGFBP-3, OR = 0.88 (0.79, 0.98); IGF-I:IGFBP-3 ratio, OR = 1.10 (0.97, 1.24). For all exposures, there was substantial heterogeneity (all $I^2 > 75\%$), partly explained by study design: the magnitude of associations was smaller in prospective versus retrospective studies, and for IGFBP-3 the inverse association with prostate cancer risk was seen in retrospective but not prospective studies. There was weak evidence that associations of IGF-I and IGFBP-3 with prostate cancer were stronger for advanced disease. Our meta-analysis confirms that raised circulating IGF-I is positively associated with prostate cancer risk. Associations between IGFBP-3 and prostate cancer were inconsistent, and there was little evidence for a role of IGF-II, IGFBP-1 or IGFBP-2 in prostate cancer risk.